

# Supplementary Material

## Risk-Aware Last-Mile Routing under Persistent Disruptions: Tail-Risk Planning and Selective Dynamic Recourse

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### Scope and provenance

Figures S1–S6 are reconstructed only from the confirmatory R3/R4/R5 result summaries carried into the revision package. No R2, smoke-test, pilot, or other legacy result is used as scientific evidence. No geographic route map is included and no Athens coordinates are inferred or invented. If a future route visualization is required in the absence of verified coordinates, it must use a two-dimensional embedding of the observed distance matrix and be labelled explicitly as a *distance-preserving schematic layout*, not a geographic map.

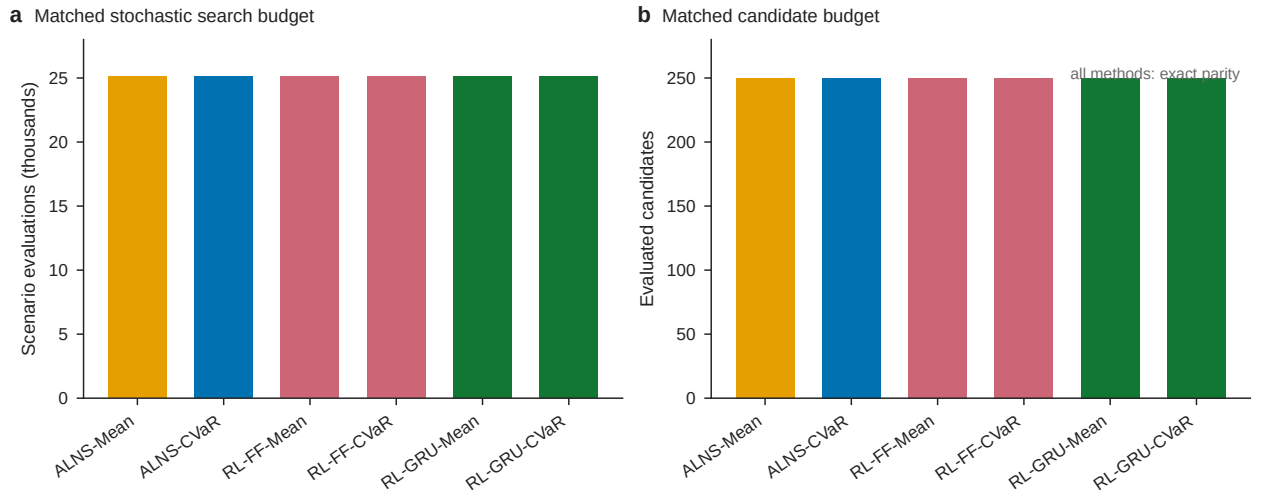


Figure 1: **R3 online-search fairness audit.** All six confirmatory source-day methods use exactly 25,100 stochastic scenario evaluations and 250 evaluated candidate routes per search. Offline RL training is intentionally excluded from this online-budget equality and is reported separately in the main text.

### Confirmatory interpretation constraints

The supplementary figures are interpreted under the same frozen scientific constraints as the main manuscript: the Hawkes process is exogenous; the OMI is a common environmental state; routes do not feed back into Hawkes intensity;  $\alpha = 0.95$  for CVaR; ALNS is genuinely adaptive; RL shows no robust incremental advantage under the matched protocol; and DDDAS conditions only on information observed before the decision epoch and edits only the affected unexecuted suffix.

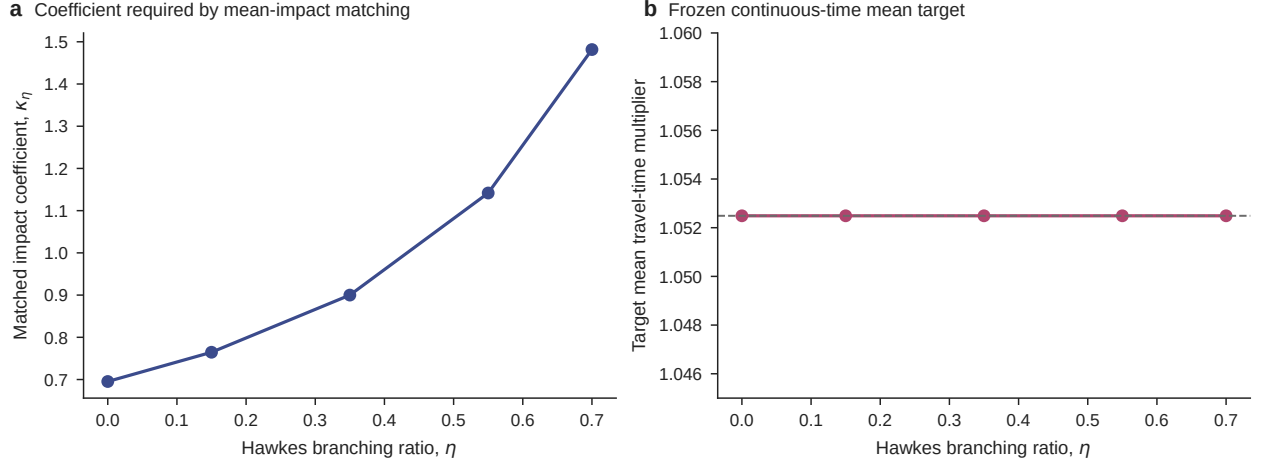


Figure 2: **R4D mean-impact matching.** The impact coefficient  $\kappa_\eta$  is adjusted as Hawkes persistence changes so that the continuous-time mean environmental travel-time multiplier is held at the frozen target 1.052489. This control separates temporal clustering from a simple increase in mean environmental impact.

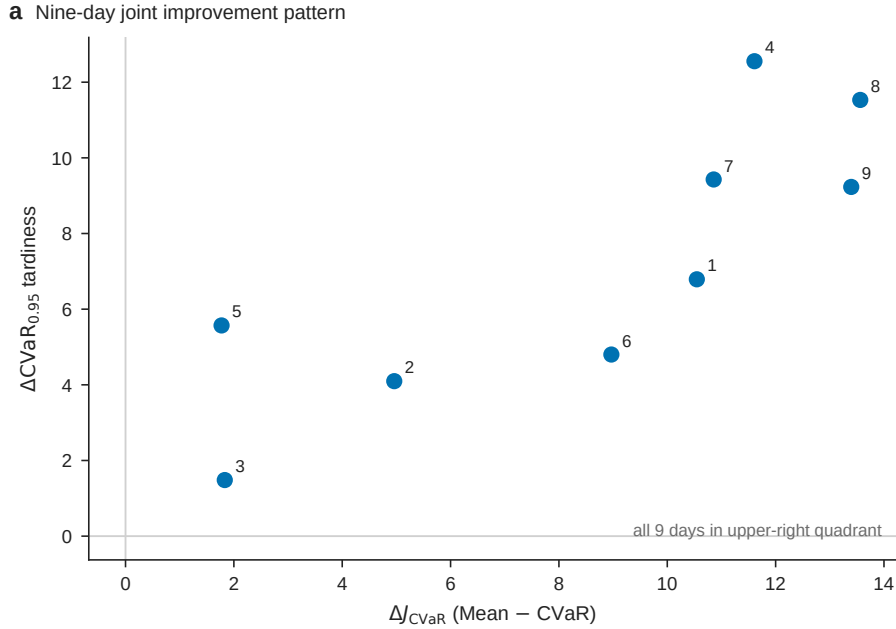


Figure 3: **R4E1 joint nine-day gains.** Each point is one operational day after averaging its three paired optimization repetitions. Both prespecified static-minus-CVaR contrasts are positive on all nine days: the aligned  $J_{\text{CVaR}}$  objective and  $\text{CVaR}_{0.95}$  tardiness. The day, rather than the optimization replicate, is the primary inferential unit.

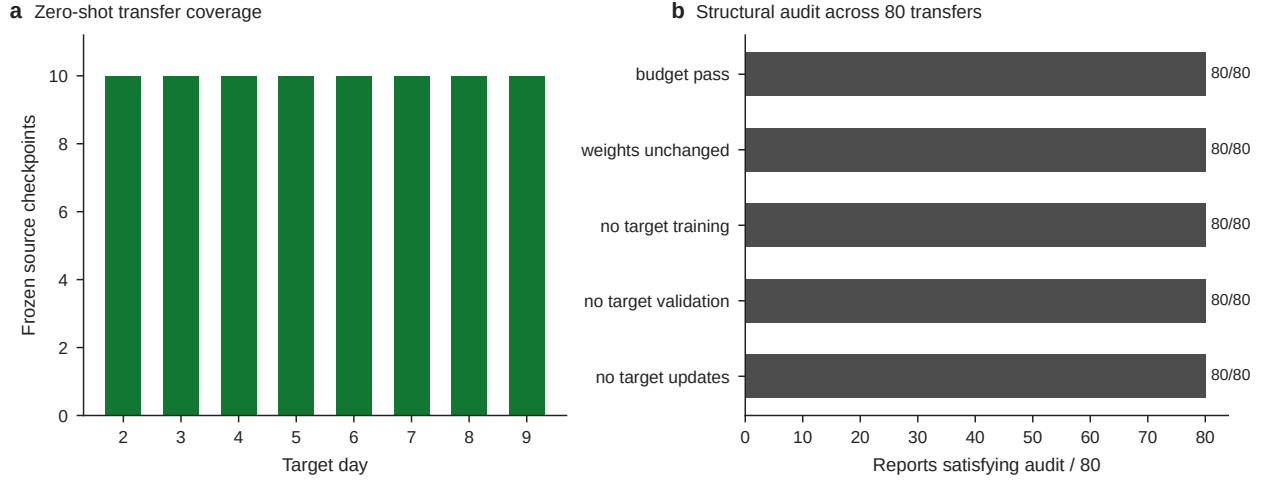


Figure 4: **R4E2 zero-shot transfer integrity.** Ten frozen Day 1 source checkpoints are evaluated on each target Day 2–9, giving 80 transfer reports. The structural audit confirms matched search budgets, unchanged learned weights, and no target-day training, validation, or gradient updates in all 80 reports. These checks support the zero-shot interpretation; they do not imply a performance advantage for RL, and the main text reports that none of the four prespecified learned-versus-uniform contrasts remains significant after Holm correction.

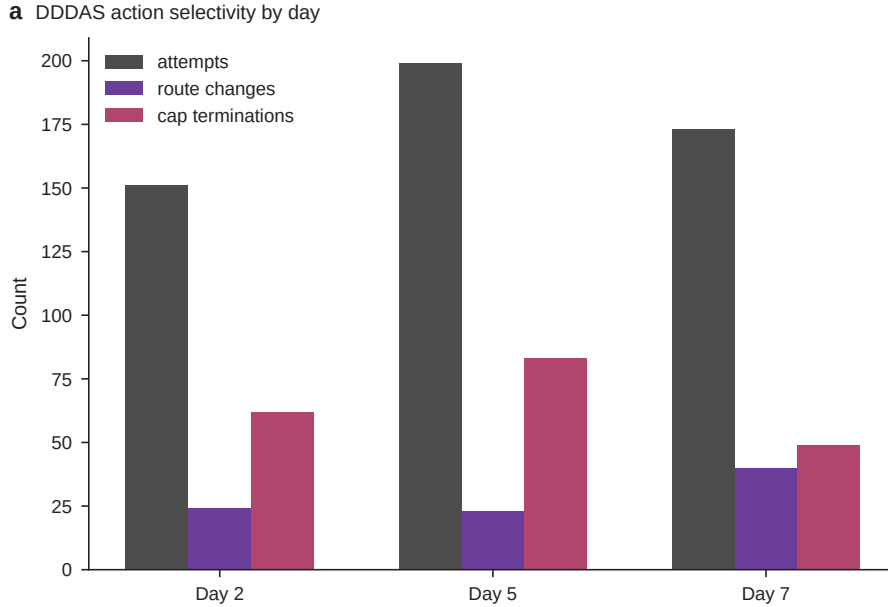


Figure 5: **R5-H bounded-recourse accounting.** Across Days 2, 5, and 7 the causal DDDAS controller makes 523 recourse attempts, executes 87 route changes, and reaches the 25,000-operator-decision cap in 194 attempts. The trigger therefore initiates reconsideration rather than compulsory rerouting.

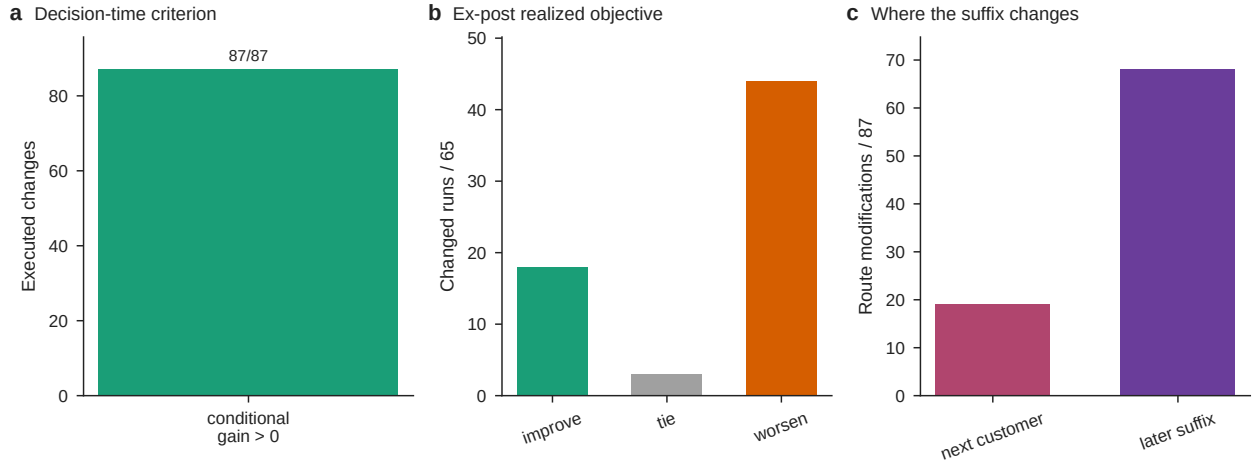


Figure 6: **R5-H conditional-versus-realized distinction.** All 87 executed route modifications improve the local conditional CVaR search objective on the decision-time scenario bank. Among the 65 complete runs with at least one route change, however, the realized composite objective improves in 18 runs, ties within  $10^{-12}$  in 3, and worsens in 44. Only 19 of 87 modifications change the immediate next customer; 68 alter a later part of the unexecuted suffix.